CTU training solutions | Private Tertiary Education

CAPSTONE PROJECT

TOPIC PROPOSAL & ACCEPTANCE

Student / Group Template

Project Title	Customer Support Ticketing System: Design and Development of a Web-Based Support Request Management Application
Group Members	Freek Schaefer-Gerber; Wian Sandenbergh
Module	Software Engineering Capstone Project
Lecturer / Supervisor	
Week / Stage	Proposal and initial research
Submission Date	29 October 2026

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Proposed Project Title

Customer Support Ticketing System: Design and Development of a Web-Based Support Request Management Application

Problem / Need Identified

Small organisations and service teams may receive customer enquiries, complaints and service requests through email, telephone calls and informal messaging channels. When these interactions are not recorded in one shared system, requests can be overlooked, duplicated or handled without clear ownership. Customers may have limited visibility into progress, while support staff may struggle to prioritise cases and report on performance. This project addresses the need for a structured, traceable support workflow; the specific organisational context will be validated during requirements gathering.

Proposed Solution

Design and develop a responsive web-based customer support ticketing application. Customers will be able to register, sign in, create tickets and view their progress. Support agents will be able to review assigned tickets, respond to customers and update ticket status. Administrators will manage users, categories, priorities and assignments and view summary reports. A relational database will store users, tickets, comments and status history, with role-based access controls and input validation.

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Expected Users / Beneficiaries

Customers who submit requests; support agents who investigate and resolve them; support administrators who coordinate work; and small organisations seeking clearer service tracking. The prototype will be demonstrated using realistic fictional users and support cases.

Initial Scope

Included: account login and role-based permissions; ticket submission and unique ticket reference; categories and priorities; ticket assignment; status tracking (Open, In Progress, Resolved, Closed); threaded comments; searchable ticket lists; a basic dashboard; relational database; and functional testing. Excluded from the initial release: paid subscriptions, AI chatbots, telephone integration, advanced service-level automation, and live payment processing. Email notifications and file attachments are optional enhancements subject to time and security review.

Supervisor Decision

☐ Approved ☐ Approved with changes ☐ Resubmit

Supervisor Comments

Provide feedback or conditions for approval.

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Sign-off

Student/Group Representative: _____ Supervisor: _____ Date: _____